Statistical Inference & Hypothesis Testing Report

Analyst: Portfolio Toolkit | **Date:** June 02, 2026 | **Framework:** AP Statistics Standard

1. State

Define the hypotheses and significance level to evaluate the claims of the A/B test.

Significance Level (α): 0.05

Hypotheses (Symbols):

$$H_0 : F_A(x) = F_B(x) \quad H_a : F_A(x) \neq F_B(x)$$

Null Hypothesis (H_0): The distributions of both groups are identical.
Alternative Hypothesis (H_a): There is a systematic difference in values between the two groups.

2. Plan

Identify the appropriate statistical procedure and verify its mathematical assumptions.

Recommended Procedure: Mann-Whitney U Test (Non-parametric)

Assumption / Condition Checked	Status & Diagnostic Details
Randomness	Assuming independent random sampling/assignment of groups.
Independence	Samples are mutually independent and represent < 10% of population.
Normality (Shapiro-Wilk)	Group A Normality (p = 0.1560, Passed). Group B Normality (p = 0.0359, Failed). Normality assumptions NOT fully met.
Equal Variance (Levene's)	Levene's Test for equal variance: p = 0.0023 (Unequal variances assumed).

■■ ASSUMPTION VIOLATION ENCOUNTERED

Some statistical conditions are violated. The tool automatically routed/recommends using Mann-Whitney U Test (Non-parametric alternative) to ensure robust inference.

3. Do

Calculate sample metrics, standard error, test statistic, and the resulting p-value.

Sample Descriptive Statistics:

Group	Sample Size (n)	Median Value
Group A	18	9.4044
Group B	20	18.7935

Calculations & Mathematical Formulations:

• *U Stat*:

$$U_{statistic} = 238.0$$

Statistical Parameter	Value
Test Statistic (U)	238.00000
p-value	0.09276

4. Conclude

State the final decision and offer a business interpretation based on the statistical results.

Compare p-value to α: 0.09276 >= 0.05. Since the p-value is greater than or equal to the significance level, we **fail to reject the null hypothesis (H_0)**. There is insufficient evidence to conclude there is a significant difference between the variations.

■ **DECISION: FAIL TO REJECT NULL HYPOTHESIS**

Business Action Recommendation: There is no statistically detectable difference in performance. The variations perform identically on average. We recommend choosing the option that has lower maintenance costs or better UX.